



Congenital diaphragmatic hernia repair in patients on extracorporeal membrane oxygenation: How early can we repair?☆☆

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ABSTRACT

Background: The benefits to early repair (<72 h postcannulation) of infants with congenital diaphragmatic hernia (CDH) on extracorporeal membrane oxygenation (ECMO) are increasingly recognized. Yet it is not known if even earlier repair (<24 h) results in comparable or improved patient outcomes. The goal of this study was to compare “super-early” (<24 h) to early repair (24–72 h) of CDH patients on ECMO.

Methods: A retrospective review of infants with CDH placed on ECMO (2004–2017; n = 72) was performed. Data collected on the patients repaired while on ECMO within 72 h of cannulation (n = 33) included pre- and postnatal disease severity stratification variables and postnatal outcomes. Comparison groups were those patients repaired within 24 h of cannulation (n = 14) and those repaired between 24 and 72 h postcannulation (n = 19). **Results:** Patients undergoing “super-early” (<24 h) repair had an average survival of 71.4% compared to the average survival of 59.7% in the early repair group. Pre- and postnatal variables predicting disease severity were not significantly different between the groups. Mean hospital stays, ventilator days, and cannulation days were statistically similar between the groups.

Conclusions: Repair of patients with CDH patients on ECMO at less than 24 h postcannulation achieves outcomes that are comparable to those of repair between 24 and 72 h. While the present data suggest that there is not a “too early” time point for CDH repair on ECMO, larger multicenter studies are needed to validate our findings and determine the overall benefits.

Type of study: Retrospective comparative study.

Level of evidence: Level III.

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Congenital diaphragmatic hernia (CDH) is a significant structural birth defect with a prevalence of approximately 1 in 3000 live births [1,2]. These patients present within a spectrum of disease severity and consequently experience a wide range of outcomes. Prognosis is closely linked to the degree of pulmonary hypoplasia and pulmonary hypertension in affected infants [1–3]. Improvements in neonatal care, such as

enhanced ventilator strategies and improved antipulmonary hypertensive therapies, have led to better overall survival rates. Even with these advances, a significant number of patients require extracorporeal membrane oxygenation (ECMO) support in the initial phase of their care [3–8].

Despite the long-standing use of ECMO as a life-saving supportive measure in infants with CDH, the current protocols to treat and manage these patients remain controversial [8,9]. One such issue is the timing of the repair of the diaphragmatic defect, which has undergone significant evolution over the past 30 years [8,10]. Initially, most institutions performed the diaphragmatic hernia repair either at the conclusion of the patient's ECMO run or in the postdecannulation period [2,3,11–14]. The Boston group was the first to report an early repair strategy, generally defined as surgery less than 72 h after cannulation [15,16]. This practice was based on the concept that the reallocation of the abdominal contents from the thoracic cavity would improve cardiopulmonary function, such that the patient would be in a better physiologic

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☆☆ How this paper will improve care: This study demonstrates no difference in outcomes in CDH patients on ECMO who are repaired <24 h of cannulation compared to patients repaired <72 h. These findings suggest that very early repair can be considered in CDH patients on ECMO.

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condition to recover from the stress of surgical repair and would derive greater benefit from the ECMO support, perhaps even optimizing the patient for decannulation [2,4,17,18].

In 2009 our institution adopted an evidence-based protocol for CDH management, wherein all infants placed on ECMO were repaired within 72 h of cannulation. We have previously reported multiple benefits to this early repair strategy, including decreased ECMO duration and a trend towards increased survival [17]. Initially, we allowed a short postcannulation time period for the infant to “adapt” to the circuit and to achieve an optimal state of anticoagulation. This period has become progressively shorter and we have performed repairs at less than 24 h postcannulation. However, it has not been reported whether a “super-early” repair within the first 24 h postcannulation is equivalent to or associated with any added benefit to repair between 24 and 72 h. Based on our collective experience, we hypothesized that “super-early” repair of the CDH defect within 24 h postcannulation could attain the same level of safety as early repair between 24 and 72 h, with comparable or potentially better results. To test this hypothesis, we performed a retrospective review of all CDH patients placed on ECMO at our institution, comparing patient outcomes between “super-early” and early repair approaches.

1. Methods

1.1. Patient population

This study is a retrospective review of data that have been prospectively collected in accordance with all Health Insurance Portability and Accountability Act (HIPAA) regulations and approval by the Baylor College of Medicine Institutional Review Board (H-26176). Accuracy of data collection was confirmed by at least two independent investigators. We reviewed the records of 72 inborn and outborn infants with the diagnosis of congenital diaphragmatic hernia and that required the use of ECMO at any point during their admission at Texas Children's Hospital (TCH) between March 2004 and March 2017. Of those 72 subjects, 14 patients underwent repair after ECMO decannulation, 5 patients died on ECMO without repair, and 2 patients required ECMO after their repair. Of the 51 patients who underwent repair while on ECMO (70.8% of all CDH patients receiving ECMO support), 18 patients were excluded because repairs were performed outside the time frame of interest (>72 h). This resulted in 33 patients who underwent repair within 72 h of ECMO cannulation, where the earliest date of repair was May 2007 and the latest was March 2017.

1.2. Study design

In our institution, the study team included four surgeons (OOO, SGK, TCL, AMV) who repair all neonatal CDH patients and who are part of the ECMO surgery team at TCH. Decisions regarding ECMO initiation and management were made among the ECMO/CDH surgeon(s) and the ECMO neonatologist (CJF). This decision-making process is captured in the publicly available TCH clinical standards report for the acute management of CDH (current as of December 2017), which includes an evidence-based treatment escalation algorithm addressing pre-ECMO therapy and indications for ECMO [19]. As in most centers, ECMO is only initiated in the presence of a reversible condition and after a failure of intensive medical management, which at TCH can include inhaled nitric oxide (iNO) and high frequency oscillation ventilation (HFOV). Once these criteria for disease severity have been met, some established clinical indications for ECMO include an oxygenation index (OI) >40 on two separate measurements, PO₂ persistently <40 mmHg, or lactate rising above 3, although each patient is managed on a case-by-case basis.

The patient groups stratified for comparison were repair on day 1 (<24 h) after cannulation and repair on days 2–3 (24–72 h) after cannulation. Measures of disease severity collected for comparison included: absolute and observed/expected total fetal lung volume (|TLV| and o/e

TLV) at the time of first magnetic resonance imaging (MRI), observed/expected lung-to-head ratio (o/e LHR) at first MRI, the presence of liver herniation, percent “liver-up,” the need for patch repair, gestational age, birthweight, Apgar score at 1 min and 5 min of life (not assigned if EXIT-to-ECMO), ECMO mode, and need for emergency C-section. If only one measurement were available (e.g., the imaging report only stated the TLV, not o/e TLV), the calculation for the expected fetal lung volume was derived using the formula put forth by Mahieu-Caputo et al.: fetal lung volume (mL) = exp. (1.24722 + 0.08939 × gestational age in weeks) [20]. Outcomes assessed included patient survival, number of days on ECMO, length of hospital stay (if survived to discharge), and number of days on mechanical ventilation (if survived to discharge without requiring home ventilator support). The Congenital Diaphragmatic Hernia Study Group (CDHSG) predicted survival, a common measure of risk stratification using a logistic regression equation based on birth weight and 5-min Apgar scores, was retrospectively calculated from <http://nicutools.org> [3].

1.3. Statistics

Statistical analyses were performed using Excel (2013, Chicago, IL) and/or SAS version 9.4 (SAS Institute, Inc., Cary, NC) by an independent biostatistician who was blinded to the study groups. Continuous variables are presented as mean ± standard deviation. Categorical variables are presented as number and percentage. Univariate comparisons were performed using Pearson's chi-square test, Fisher's exact test, or non-parametric Wilcoxon rank sum test, as appropriate. To identify independent predictors, we built multivariable logistic regression models using all clinically relevant factors and a backward selection method with a removal p-value of 0.15. Model fit was assessed by Hosmer and Lemeshow goodness-of-fit test and ROC curve. A p-value <0.05 was considered statistically significant.

2. Results

2.1. Patient demographics and pre-operative variables

Of 72 total patients with CDH that received ECMO support at any point during their hospital admission during 2004–2017, 33 patients underwent repair within 72 h of ECMO cannulation (45.8% of all CDH patients receiving ECMO support). Of those 33 patients, 14 (42.4%) had “super-early” repair (<24 h of cannulation) and 19 patients (57.6%) underwent early repair (between 24 and 72 h of cannulation). One patient in the early group and 2 patients in the super-early group were outborn, and thus selected prenatal imaging data were unavailable. Only 2 patients (early group) were not prenatally diagnosed, and in 1 patient (early group), fetal MRI could not be obtained owing to maternal size and access to care. Two patients (super-early group) that underwent EXIT-to-ECMO did not have Apgar scores assigned. No patients included in this study underwent tracheal occlusion (TO).

The demographics and prenatal characteristics of the two cohorts are presented in Table 1. Overall, the group of patients repaired at <24 h were well-matched to those repaired between 24 and 72 h according to prenatal risk stratification predictors. Super-early and early repair groups were not significantly different with respect to gender (71.4% vs 78.9% male, respectively; p-value = 0.63) or hernia sidedness (73.7% vs 64.2% left-sided, respectively; p-value = 0.58). No patients had bilateral hernias. There was also no significant difference between the super-early and early groups as to CDH associated with other major anomalies (50% vs 57.9% nonisolated CDH, respectively; p-value = 0.53), including Fryn's syndrome, Russell–Silver syndrome, chromosomal microdeletions or gains, or various cardiac abnormalities (atrial septal defect [ASD], ventricular septal defect [VSD], hypoplastic aortic arch, Tetralogy of Fallot, etc.). The average gestational age of patients in both groups at the time of their first MRI or ultrasound was 25.9 ± 4.6 weeks. Absolute total lung volumes (7.8 vs 11.1 mL,

Table 1
Patient demographics.

Variable	Super-early repair (n = 14)	Early repair (n = 19)	p-value	All patients (n = 33)
Male	71.4%	78.9%	0.63	75.8%
Left sided CDH	64.3%	73.7%	0.58	69.7%
Isolated disease	50%	42.1%	0.53	45.5%
Cardiac anomaly	42.9%	36.8%		39.4%
Abnormal karyotype	14.3%	15.9%		15.2%
Mean gestational age at birth (weeks)	37.4 (± 2.0)	38.1 (± 1.39)	0.29	37.8 (± 1.7)
Mean birthweight (g)	2803.2 (± 633.5)	2830.4 (± 529.7)	0.89	2818.9 (± 566.6)
1st O/E LHR (mean) ^a	37.5% (± 9)	32.2% (± 14)	0.28	34.5% (± 13)
1st O/E TLV (mean) ^b	22.8% (± 7.0)	24.6% (± 11.4)	0.63	23.8% (± 9.5)
Absolute TLV (mL or cm ³) ^b	7.8 (± 6.0)	11.0 (± 5.8)	0.18	9.56 (± 6.0)
Intrathoracic liver position	100%	100%	-	100%
Percent liver herniation ^b	31.3% (± 11.9)	25.3% (± 9.9)	0.16	28% (± 11.1)
Patch repair	92.9%	100%	0.34	97.0%
Apgar 1 min ^c	3.25 (± 1.8)	4 (± 2.0)	0.29	3.7 (± 1.9)
Apgar 5 min ^c	6.9 (± 1.4)	6.8 (± 1.6)	0.82	6.8 (± 1.5)
Emergency C-section	50%	21.0%	0.09	32.3%
VA ECMO	64.3%	73.7%	0.58	69.7%

^a Data unavailable in 5 patients; outborn or not prenatally diagnosed.^b Data unavailable on 6 patients; outborn, not prenatally diagnosed, or no fetal MRI.^c Data unavailable on 2 patients; Apgar not assigned if EXIT-to-ECMO.

respectively; p-value = 0.18), and observed/expected total lung volumes (22.8% vs 24.5%, respectively; p-value = 0.63) were not significantly different between the super-early and early groups, nor were observed/expected lung-to-head ratios (36% vs 33%, respectively; p-value = 0.28). All patients had liver herniation into the chest and the percent liver herniation (“liver up”) was not significantly different between the super-early and early groups (31.3 vs. 24.8%, respectively; p-value = 0.16). All patients were repaired via an open abdominal approach and all but 1 patient in the super-early group required patch repair.

In comparing peri- and postnatal risk factors, the two groups were once again statistically similar. There was no significant difference between the super-early repair and early repair groups with respect to mean birthweight (2803.2 g vs 2830.4 g, respectively; p-value = 0.89) or mean gestational age at birth (37.4 vs 38.1 weeks, respectively; p-value = 0.29). Prematurity of 34 weeks or less affected only one patient in each group (2 of 33 patients, 6%). The super-early and early repair groups did not have statistically different mean Apgar scores at 1 min of life (3.25 vs 4, respectively; p-value = 0.29) or at 5 min of life (6.9 vs 6.8, respectively; p-value = 0.82). Two patients in the super-early group underwent EXIT-to-ECMO and as such were not assigned Apgar scores. The super-early repair patients trended towards a higher likelihood of needing emergent C-section (42.9% vs 21.1%; p-value = 0.09).

26 of the 33 study patients were initiated on ECMO within the first 24 h of life (on average, at 17.3 ± 20 h of life); one outborn patient (early group) was cannulated at day-of-life 18 and underwent repair 3 days thereafter. The ECMO mode most commonly chosen in both groups was venoarterial (VA), with 64.3% of the super-early group and 73.7% of the early repair group on this mode; both of the EXIT-to-ECMO patients were placed on VA mode. Notably these values are somewhat lower than the percentage of CDH patients on VA ECMO cited in the recent literature [4,20,21]. At our institution, venovenous (VV) ECMO is offered to CDH patients without severe cardiac dysfunction and with the ability to accept a 13 Fr dual lumen VV cannula. The

two EXIT-to-ECMO patients were both placed on VA and assigned to the super-early group. One patient in the early repair group required a conversion from VV to VA ECMO.

2.2. Patient survival and outcomes

Outcomes for the two cohorts are presented in Table 2. The overall survival of the 33 patients who underwent CDH repair while on ECMO was 63.6%, which is similar to the CDH Study Group (CDHSG) predicted survival for the whole group (62.85 ± 22.1%). Interestingly, out of all the severity stratification parameters analyzed, LASSO regularized logistic regression identified male sex and o/e TLV values over 25% as factors that possibly reduced the likelihood of death, although this relationship was not statistically significant.

Survival was 71.43% in the super-early repair group and 57.89% in the early repair group, but this difference was not statistically significant (p-value = 0.44). The CDHSG predicted survival scores for the super-early (63.2 ± 19.7%) and early groups were comparable as well (62.5 ± 21.5%). However, the super-early repair group trended towards hospital stays averaging almost 2 months shorter than those of the early repair group (115.4 vs 175.5 days if survived to discharge, respectively; p-value = 0.09). There was no difference between the super-early and early repair groups with respect to ECMO duration (217.7 vs 257.7 h on ECMO, respectively; p-value = 0.50) and days on mechanical ventilation (29.2 vs 46.5 days if survived to discharge without ventilator care, respectively; p-value = 0.18). One patient in the super-early repair group and 4 patients in the early repair group required tracheostomy and home mechanical ventilatory support.

2.3. Comparison to previous eras of institutional management

In 2009, our institution protocolized early repair (<72 h) of CDH on ECMO. To see if there was a difference in survival after this change in management, the data were divided into two eras: pre-early-repair

Table 2
Outcomes.

Variable	Super-early repair (n = 14)	Early repair (n = 19)	p-value	All patients (n = 33)
Survival	71.4%	57.9%	0.44	63.6%
Time on mechanical ventilation (days) ^a	29.2 (± 18.0)	57.25 (± 51.6)	0.18	42.4 (± 39.1)
Duration of ECMO therapy (hours)	217.7 (± 180.2)	257.7 (± 149.6)	0.50	240.7 (± 161.8)
Length of stay (days) ^a	115 (± 79.6)	175.5 (± 73.3)	0.09	146.9 (± 80.5)

^a If survived to discharge.

protocol (2007–2009) and post-early-repair protocol (2009–2017). Only 4 patients underwent repair before 72 h in the pre-protocol-change era, including one repaired <24 h of cannulation, and survival was 100%. A total of 29 patients underwent repair before 72 h in the post-protocol-change era, of which 13 (44.8%) were <24 h of cannulation and 16 (55.1%) were between 24 and 72 h of cannulation. The overall survival rate was 58.6%, with the group undergoing repair within 24 h of cannulation having a 69.2% survival rate and the group undergoing repair between 24 and 72 h of cannulation having a 50% survival. Overall, there was no evidence that the year of surgery affected survival rates.

A publication out of this institution in 2013 reported on the benefits of early repair of CDH on ECMO [17], showing a 73% survival in patients repaired <72 h between 2004 and 2011. In the era of overlap between our studies (2007–2011), 10 patients underwent repair within 72 h of cannulation, with an overall survival rate of 70%. Of those 10 patients, 4 patients (40%) had repair <24 h of cannulation with a survival rate of 75%. The remaining 6 patients that underwent repair between 24 and 72 h had a survival rate of 66.7%. The results from this era of overlap are thus largely congruent with the overall survival rate seen in the 2013 paper (73%) and the differential survival rates seen in our manuscript (<24 h: 71%; between 24 and 72 h: 58%).

3. Discussion

In this study, we found that the outcomes of patients undergoing “super-early” repair of CDH on ECMO (<24 h of cannulation) were similar to those of their early repair (between 24 and 72 h after cannulation) counterparts. To our knowledge, this is the first study to validate that diaphragmatic repair within 24 h of ECMO cannulation does not diminish the benefits of early repair that have been previously reported. The overall survival of our 33 patients who underwent CDH repair within 72 h of ECMO cannulation was comparable or superior to the survival rates of CDH patients on ECMO reported in the literature [2,11,17,18,22,23], and our super-early group demonstrated a trend towards shorter hospital stays, although the difference was not statistically significant. Overall, super-early repair does not seem to result in a disadvantage to patient outcomes and survival compared to early repair.

The principle behind early repair of CDH on ECMO is to hasten an improvement in cardiopulmonary physiology by decreasing the mass effect of the herniated abdominal contents. It is anatomically and physiologically plausible to suggest that normalizing thoracic anatomy – allowing the lungs to maximally expand, the heart to correctly position, and relief of bronchial compression – would address some of the root causes of these patients' physiologic instability, primarily impaired venous return, cardiac dysfunction, and hypoxemia. Furthermore, early repair provides the benefit of a full ECMO run for postoperative support and may permit a less technically challenging operation by avoiding the onset of inflammation and anasarca, which occurs after the initiation of ECMO [18]. In our and other institutions, early repair has been associated with significantly reduced morbidity and higher survival rates than those of the CDH registry patients [17,18]. This study specifically addresses the timing of these surgeries, and our findings suggest that it may be reasonable to expect that super-early repair has the potential to optimize the anatomic and physiologic advantages provided by early repair.

Concern for operative bleeding and embolic phenomena limits enthusiasm for any surgical intervention while on ECMO, as significant bleeding has been shown to be a hallmark of poor outcomes [6,24,25]. Multiple authors have reported strategies to mitigate these complications, including the use of aminocaproic acid (Amicar), tranexamic acid (TXA/TEA), topical thrombin on suture lines, minimizing the dissection of the posterolateral diaphragmatic rims, heparin-free surgery, and lower ACT thresholds [2,4,26–30]. With the application of these strategies, differences in outcomes and even in bleeding complications between patients repaired on ECMO and post-ECMO have decreased [31]. At our institution, pediatric surgeons, neonatologists, and transfusion medicine specialists collaboratively manage systemic anticoagulation for

CDH patients on ECMO within an institutional protocol, which has evolved over the duration of the current study. Currently, heparin, the primary systemic anticoagulant, is titrated to meet coagulation and hemostatic needs. This is achieved primarily by using serial anti-Xa activity along with PT, aPTT (with and without hepzyme), fibrinogen, D-dimer, platelet count, and antithrombin activity. This protocol, managed on a patient-by-patient basis, has allowed us to minimize bleeding risks and supports the possibility of repairing these neonates within 24 h of cannulation.

The rationale and benefits of CDH repair within 72 h of cannulation appear clear, but the question remains: what are the appropriate physiologic benchmarks at which early repair should be initiated? Currently, the most widely used indication for early repair is, colloquially, “stability on the circuit,” but given the interinstitutional variability in patient population, criteria for ECMO use, and overall management of CDH, it is near-impossible to universally define a specific set of criteria. Some of the more accepted criteria are stabilized anticoagulation profiles – for example, an activated clotting time (ACT) of approximately 180 s and minimal changes to the heparin dose – and the neonate being physiologically normalized. Our results suggest that, if the anecdotal criteria for steady state are met at a super-early time point, then repair at that time is appropriate as there is no benefit to delaying surgery.

The limitations to this study include its retrospective nature and single institution experience, as our protocols and predictive tools are not universally used. These data were also limited by the small sample size, thus limiting the statistical power of the study. One of our future directions is a prospective observational study that further investigates these trends with an adequately powered sample, which will enable us to perform more advanced statistical analyses to control for possible confounders.

In conclusion, the present study finds that diaphragmatic repair less than 24 h after ECMO cannulation is comparable to repair in the 48 h thereafter. Based on these results, a broader investigation is warranted to definitively determine whether “super-early” diaphragmatic repair in patients with severe CDH results in equivalent or better outcomes than traditional management.

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